

Jeremy Fontenot's On-Premises Home Lab Documentation

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Abstract

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This document presents a professional, public-safe description of Jeremy Fontenot's personal on-premises home lab. It is written for recruiters, hiring managers, and technical reviewers who need a clear view of the environment, the evidence behind the claims, and the boundaries around what is not being claimed.

The lab supports systems administration and infrastructure operations practice while Jeremy's professional positioning remains anchored in experienced Service Desk and IT Support work. The evidence set demonstrates disciplined validation, documentation, troubleshooting, and automation habits without presenting personal lab activity as employer production administration.

Purpose and Scope

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The purpose of this documentation is to explain what exists in the personal lab, why each component matters, how evidence was collected, and how claims are classified. The scope includes the physical host, Proxmox virtualization, pfSense routing, Windows identity services, Linux integration, backup architecture, logging, automation, validation workflows, and known limitations.

The document intentionally avoids unsupported claims of enterprise scale, guaranteed availability, production responsibility, or full security assurance. It focuses on what a reviewer can inspect from sanitized artifacts, source summaries, manifests, hashes, and current-state validation records.

Environment Boundaries

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The environment is a personal nonproduction home lab. It is appropriate evidence for technical initiative, hands-on practice, systems thinking, and documentation discipline. It is not a claim of employer production administration, client infrastructure ownership, or operations at enterprise scale.

Public artifacts are sanitized. Private credentials, tokens, cookies, private keys, raw logs with sensitive values, and administrative secrets are not published. Historical evidence remains available only where it helps reviewers understand chronology, not as a current-state label.

Evidence Methodology

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Evidence precedence favors current command output, API output, guest-agent output, authenticated target-local inventory, and explicit repository validation. Older screenshots and documentation remain useful when clearly labeled as historical or when current records independently support the same claim.

The methodology separates observation from interpretation. A TCP timeout is reported as a timeout, not as proof of an outage. A configuration record is reported as configured unless a test or authenticated inventory supports a stronger statement.

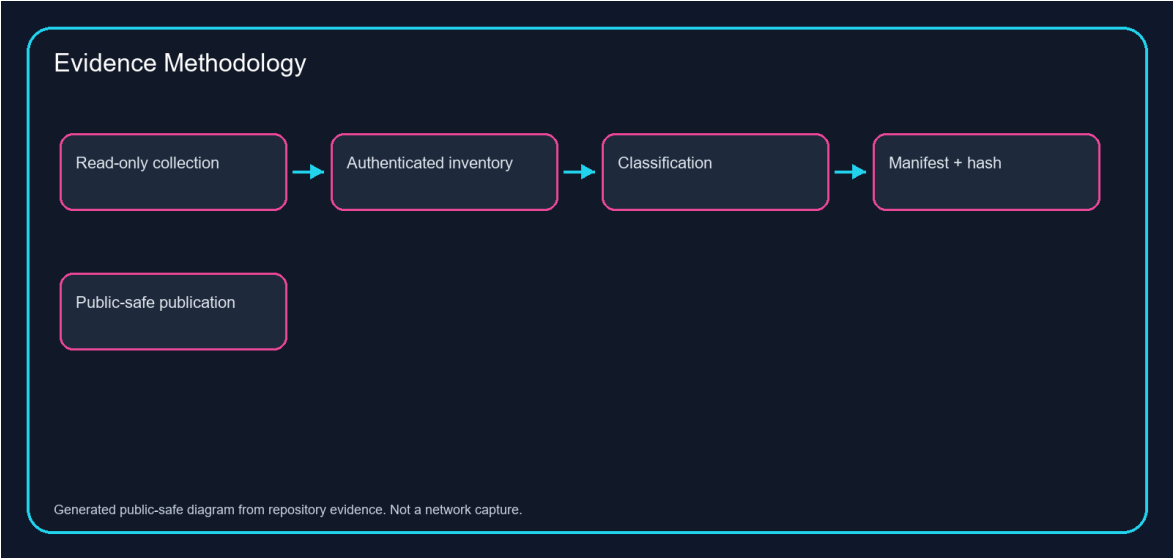


Figure 2. Evidence methodology from read-only collection to public-safe publication.

Evidence Classification

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Claims are classified as Validated, Tested, Configured, Historical, Limitation, or Inconclusive. Validated means direct evidence supports the stated behavior or state. Tested means a relevant check produced a recorded result, but the result may be narrower than the broad service claim.

This classification model helps technical reviewers trust the strongest statements while still seeing what remains unresolved. It also keeps recruiter-facing copy readable by moving dense proof details into manifests and appendices.

Table 1. Evidence classification model.

Classification	Use	Boundary
Validated	Direct evidence confirms the state or behavior.	Used only where source artifacts support the claim.
Tested	A relevant test produced a recorded result.	May be narrower than full service health.
Configured	Configuration is present.	Behavior may require a separate validation.
Historical	Useful retained context.	Not promoted as current state.

Classification	Use	Boundary
Limitation	Known gap or unsupported conclusion.	Kept visible for reviewers.

Physical Server Architecture

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The physical lab host is documented as a Dell PowerEdge R710. It provides the hardware foundation for the virtualization environment and supports the lab's role as a practical infrastructure learning platform.

The server is presented as personal lab equipment. Hardware ownership and capacity are not extended into claims about production-grade resilience, clustered high availability, or enterprise hardware lifecycle management.

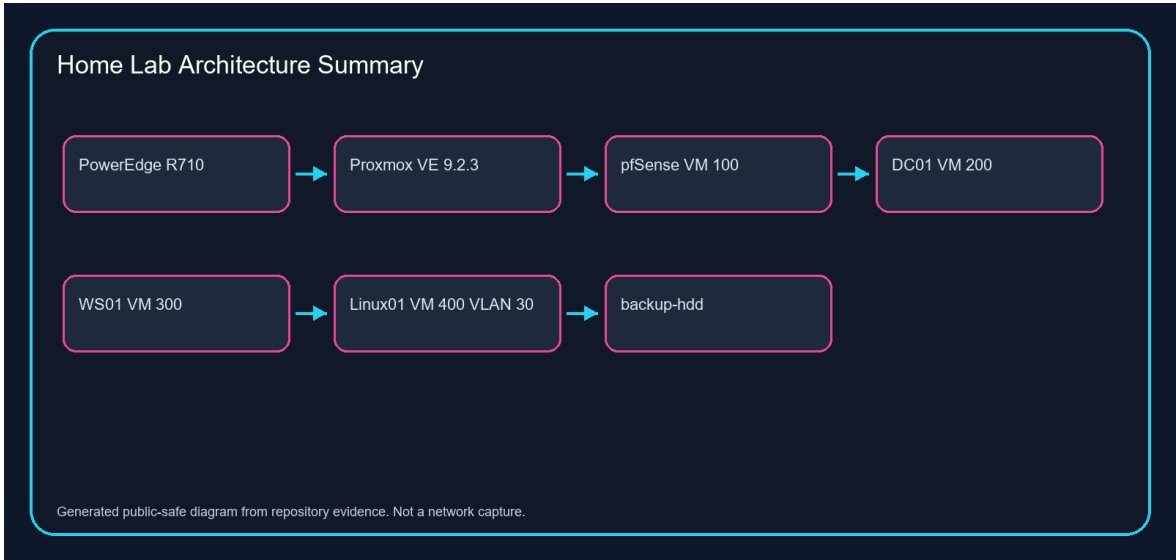


Figure 1. Current supported home-lab architecture summary.

Proxmox Virtualization

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Current source records identify Proxmox VE 9.2.3 as the virtualization platform. The primary supported virtual machines are VM 100 pfSense, VM 200 DC01, VM 300 WS01, and VM 400 Linux01.

Proxmox evidence supports running VM inventory, storage visibility, backup job visibility, Linux01 VLAN tagging, and QEMU guest agent status where available. Guest operating system claims are strengthened through target-local inventories where Proxmox metadata alone is insufficient.

Table 2. Current primary VM inventory.

VM	System	Supported role	Evidence boundary
100	pfSense	Firewall and routing VM	Configuration context; not a full firewall audit.
200	DC01	Windows Server domain controller	Guest-level claims use target-local evidence where available.
300	WS01	Windows domain workstation	Guest-agent limitation remains visible.
400	Linux01	Ubuntu domain member on VLAN 30	Guest agent and target-local evidence support stronger claims.

Storage Architecture

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Current records identify backup storage named backup-hdd at /mnt/pve/backup-hdd. Older references to other backup storage labels are treated as stale or inconclusive unless later evidence proves a rename or replacement.

Storage claims are limited to observed storage inventory, configured backup target use, and documented utilization or monitoring records. The evidence does not claim full storage redundancy or enterprise storage design.

pfSense

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pfSense is represented as VM 100 and provides routing and firewall context for the personal lab. Public documentation distinguishes pfSense configuration records from behavior that requires a separate current test.

The supported design includes a LAN gateway, a Linux VLAN gateway, and an OpenVPN management path. Direct WAN-side management exposure is treated as historical where older records mention it; the current design emphasizes VPN-scoped administration.

Network Bridges

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Proxmox bridge evidence identifies VM network attachment through vmbr interfaces. pfSense bridges between upstream and internal lab networks, while domain systems reside on the internal lab side.

Bridge documentation is used to explain how systems relate to one another, not to imply that every firewall rule, NAT rule, or packet path was exhaustively tested from the public website environment.

VLANs

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Linux01 is currently treated as VLAN 30 on vmbr1 with the supported address 10.10.30.20/24. Older Linux01 10.10.20.x evidence is preserved as historical setup or pre-reconciliation context.

VLAN claims are supported by current Proxmox metadata, DNS reconciliation, and target-local Linux inventory. The documentation avoids implying broad segmentation assurance beyond the cited evidence.

VPN and Management Routing

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The management path includes OpenVPN network context and route evidence. Current records support DNS and RDP reachability to DC01 through the management path, while SMB and other timeouts remain documented observations only.

Management routing is described as a controlled access pattern for lab administration. It is not presented as a comprehensive remote-access security audit.

Table 3. Management-path evidence boundaries.

Observation	Recorded result	Public claim
DC01 DNS TCP 53	Reachable through management path	DNS reachability supported.
DC01 RDP TCP 3389	Reachable through management path	Management RDP path supported.
SMB or policy-dependent ports	Timeouts recorded	Timeout only; not outage proof.
Direct WAN RDP	Historical, not current public design	Not promoted as current access model.

DC01

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DC01 is documented as VM 200, a Windows Server 2022 domain controller for ad.jeremyfontenot.online. Target-local inventory and retained screenshots support the Windows services that are described in this paper.

DC01 evidence matters because it anchors the lab's identity, name resolution, address assignment, policy, and file-service practice areas.

Active Directory Domain Services

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Active Directory Domain Services evidence supports a functioning personal lab domain with domain controller context, domain identity, and administrative validation records. Claims remain scoped to the personal lab.

The AD DS section demonstrates directory-services concepts: domain controller role, workstation membership, Linux domain integration, DNS dependency, and policy-linked administration.

FSMO Roles

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Retained DC01 evidence supports FSMO role ownership review. FSMO role validation is useful because it shows the domain controller role is not merely a label but part of an inspected AD DS configuration.

The evidence is not expanded into multi-domain or multi-domain-controller operational claims.

DNS

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DNS evidence includes AD-integrated DNS context, forward lookup, reverse lookup, and current Linux01 name reconciliation. DNS is central to domain services, Linux integration, and management-path validation.

Where DNS records are historical, the documentation labels them accordingly. Current Linux01 DNS resolves to 10.10.30.20.

DHCP

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DHCP evidence includes scope, option, statistics, and redacted lease review. DHCP claims are scoped to visible records and current summaries rather than broad address-management maturity.

Redaction is intentional because DHCP lease records can expose host identifiers. Published descriptions preserve technical meaning without disclosing unnecessary details.

Group Policy

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Group Policy evidence supports GPO inventory and link review. Group Policy appears in the workstation, file-service, and permissions evidence where applied policy affects access or configuration.

The documentation distinguishes inventory from processing proof. A listed GPO is not treated as applied to a user or machine unless a relevant validation record supports it.

File Services

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File-service evidence is primarily represented through DC01 and WS01 validation records, including mapped drive and share-access testing. The lab demonstrates permission modeling and user access validation.

No broad enterprise file-service assurance, data classification program, or production data management claim is made.

Security Groups and Permissions

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Security group and permission evidence includes different access outcomes for different user contexts, including read/write and write-denial validation. This demonstrates practical access-control testing.

The evidence is valuable because it shows both positive and negative validation: successful access for an authorized context and denied write access where appropriate.

WS01

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WS01 is documented as VM 300, a Windows domain workstation. Current source records include authenticated inventory evidence and historical workstation screenshots. Proxmox guest-agent limitations for WS01 remain visible where applicable.

WS01 demonstrates endpoint administration concepts: domain join, DNS/DHCP context, RDP troubleshooting, firewall state, mapped drives, and file-share permissions.

Linux01

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Linux01 is documented as VM 400, Ubuntu Server 26.04 LTS, a domain member on VLAN 30 at 10.10.30.20/24. Current Proxmox metadata and target-local inventory strengthen this state.

Linux01 provides the lab's strongest cross-platform evidence: Linux identity integration, SSSD remediation, Kerberos, SSH, UFW, sudo controls, QEMU guest agent response, and rsyslog receiver state.

Linux VLAN Placement

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Linux01 VLAN placement is supported by Proxmox VM metadata showing tag 30 and by current DNS reconciliation to 10.10.30.20. Historical 10.10.20.x references remain historical.

This section matters because correcting Linux01 placement prevents outdated network statements from leaking into public proof labels.

Active Directory Integration

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Linux Active Directory integration is documented through realm membership, identity lookup, and related validation artifacts. The public claim is that integration is configured and validated where the target-local records support it.

The integration demonstrates cross-platform administration, DNS dependency awareness, identity troubleshooting, and careful validation of authentication behavior.

SSSD

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The Linux01 SSSD remediation record documents a responder activation conflict, configuration correction, service restart, failed-unit cleanup, identity lookup, and a new AD-authenticated SSH session.

The important operational pattern is rollback-first remediation: preserve a backup, validate configuration, change only the conflicting state, restart the service, then validate identity and access.

Kerberos

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Kerberos is part of the Linux domain-integration story. The document treats Kerberos claims as supported where realm, identity, and authentication evidence connect to the Linux01 validation workflow.

Timeouts from the current management workstation to Kerberos ports are not interpreted as Kerberos service failure because the evidence path and firewall policy boundaries matter.

SSH

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SSH evidence includes post-remediation AD-authenticated access to Linux01. That result supports a meaningful operational claim: identity integration was not merely configured; it was validated with a new login path.

The documentation does not publish credentials, keys, or raw sensitive session data.

UFW

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UFW appears in Linux01 records as part of the security-control boundary. The documentation reports UFW state only to the level supported by target-local inventory and sanitized summaries.

Firewall claims are scoped narrowly because a firewall state listing is not equivalent to a comprehensive security test.

sudo Controls

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sudo controls are included as Linux administrative access evidence. The lab documentation treats sudo configuration as a privileged-access control that should be described without exposing sensitive policy internals.

The public artifact emphasizes that delegated administrative control exists where supported, but it does not publish secrets or overstate least-privilege maturity.

QEMU Guest Agent

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Current Proxmox records show Linux01 QEMU guest agent responding. Other primary VMs are not promoted to guest-agent-validated state unless the relevant current evidence supports it.

This distinction prevents hypervisor metadata from being mistaken for guest-level validation across every VM.

Centralized Logging

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Centralized logging evidence includes Linux01 rsyslog receiver state from authenticated target-local inventory and planning records for further expansion. pfSense forwarding is documented in retained records.

Additional Windows Event Forwarding, Proxmox logging, alerting, and firewall changes remain planned until explicitly approved and validated.

Backup Architecture

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Backup architecture evidence supports an enabled scheduled backup job named backup-critical-lab-vm for VMs 100, 200, 300, and 400 using snapshot mode, zstd compression, and backup-hdd storage.

This evidence supports backup configuration visibility. It does not prove current archive integrity or recurring recovery success by itself.

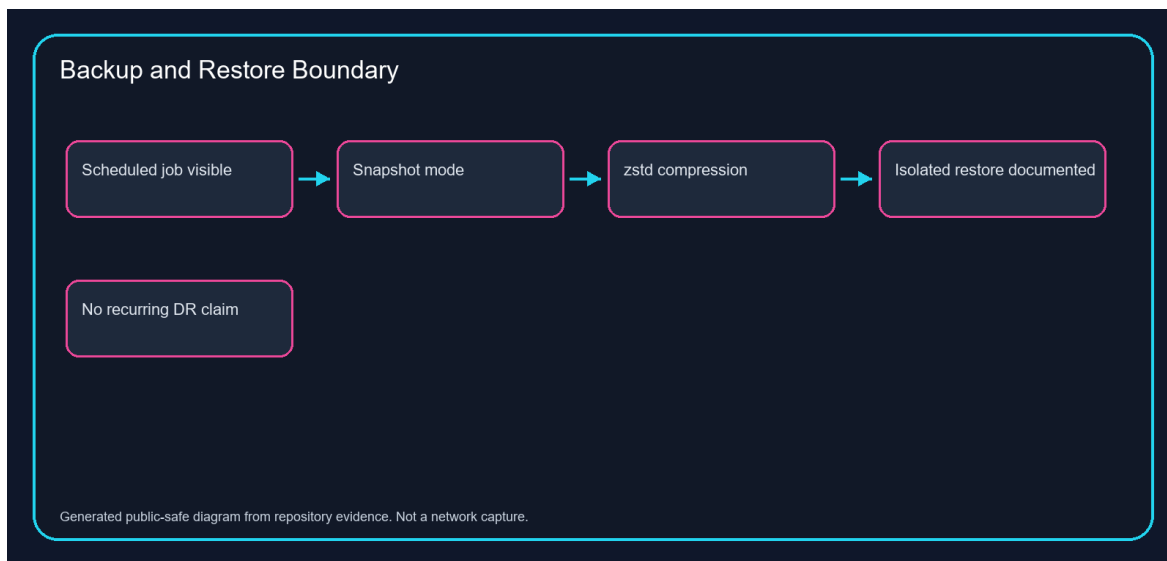


Figure 3. Backup architecture and restore-claim boundary.

Table 4. Backup architecture evidence.

Item	Supported value	Boundary
Backup job	backup-critical-lab-vm	Configured and visible.
VM coverage	100, 200, 300, 400	Does not prove restore success.
Mode	snapshot	Configuration evidence only.
Compression	zstd	Configuration evidence only.
Storage	backup-hdd	Current observed storage target.

Snapshots

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Snapshot evidence appears in workstation baseline and backup documentation. Snapshots are treated as operational checkpoints and backup-mode context rather than a substitute for tested recovery.

The document keeps snapshot configuration separate from restore assurance.

Archive Integrity

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Archive integrity evidence includes SHA-256 hash records for published documentation and retained evidence packages. Hashes help reviewers verify that artifacts are stable and traceable.

A hash proves file identity, not truth of every claim inside the file. The manifest therefore pairs hashes with claim boundaries and source paths.

Restore Testing

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Retained documentation records isolated startup restore checks for temporary WS01 and Linux01 restore VMs, with network isolation and cleanup noted. Those temporary VMs are no longer present in current Proxmox inventory.

The current records do not establish recurring disaster recovery assurance, RTO, RPO, or a standing recovery program. A future restore test would need fresh evidence with selected backup, restore target, isolation method, validation checks, and cleanup.

Storage Monitoring

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Storage monitoring claims are limited to available inventory, utilization, and planning records. Backup storage is visible, but full monitoring maturity is not claimed.

The logging expansion plan identifies disk-use thresholds and rotation concerns for Linux01, which demonstrates operations planning without claiming an implemented alerting platform.

PowerShell Automation

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PowerShell automation appears in repository validation, read-only baseline tooling, evidence hash generation, link checks, and target-local Windows inventory scripts. It demonstrates repeatable operations work rather than one-off screenshots.

The automation principle is consistent: collect read-only evidence, write structured outputs, avoid secrets, and validate repository quality before publication.

Read-Only Operations Framework

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The read-only framework includes Proxmox inventory collection, route evidence, DNS checks, bounded TCP reachability checks, repository validation, and source reconciliation. It is deliberately constrained to avoid unauthorized infrastructure changes.

Read-only validation is useful because it creates repeatable evidence while preserving production-safe habits: explicit scope, no secret capture, no configuration changes, and clear timeout wording.

Authenticated Target-Local Inventories

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Authenticated target-local inventories strengthen guest claims for DC01, WS01, and Linux01. These records provide more reliable guest-level evidence than external reachability alone.

Target-local collection is especially important where remote client context lacks modules, firewall access, or guest-agent response.

Troubleshooting and Remediation

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The strongest remediation example is Linux01 SSSD. It shows diagnosis, backup, configuration correction, service recovery, failed-state cleanup, identity lookup, and AD-authenticated SSH validation.

WS01 screenshots also support troubleshooting practice around RDP, NLA, DNS, DHCP, firewall rules, and workstation identity. These are presented as support and lab operations evidence, not production incident claims.

Known Limitations

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Known limitations include personal lab scope, no employer production administration claim, no client infrastructure ownership, no enterprise scale claim, no guaranteed uptime, no full security assurance, and no recurring restore assurance.

Timeout observations remain timeout observations. Historical screenshots remain useful only when labeled and scoped correctly.

Table 5. Non-negotiable public claim boundaries.

Area	Not claimed	Reason
Employment	Employer production administration	Personal lab evidence is separate from work history.
Scale	Enterprise-scale operations	Single personal lab environment.
Availability	Guaranteed uptime	No production SLA evidence.
Security	Full security assurance	Configuration and tests are scoped.
Recovery	Recurring DR, RTO, or RPO	Only retained isolated restore checks are documented.

Skills Demonstrated

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The lab demonstrates service-desk-to-infrastructure growth: troubleshooting, documentation, Active Directory fundamentals, DNS and DHCP awareness, Windows workstation administration, Linux administration, PowerShell automation, evidence governance, and cautious public communication.

The strongest professional signal is not a single tool. It is the habit of connecting claims to artifacts, stating limitations plainly, and validating before publishing.

Conclusion

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Jeremy Fontenot's on-premises home lab is a credible personal infrastructure portfolio because it is documented with current-state evidence, carefully bounded public claims, and reviewer-accessible artifacts.

The environment supports a clear career narrative: experienced Service Desk professional actively building evidence and validation toward Systems Administration and Infrastructure Operations.

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Appendices

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Appendix A lists the primary source artifacts. Appendix B summarizes claim boundaries. Appendix C maps required topics to document sections. Appendix D records the clickable TOC validation results.

Table 6. Required section-coverage matrix.

Topic	Document location
Title page	Title page
Abstract	Abstract
Purpose and scope	Purpose and scope
Environment boundaries	Environment boundaries
Evidence methodology	Evidence methodology
Evidence classification	Evidence classification
Physical server architecture	Physical server architecture
Proxmox virtualization	Proxmox virtualization
Storage architecture	Storage architecture
pfSense	pfSense
Network bridges	Network bridges
VLANs	VLANs
VPN and management routing	VPN and management routing
DC01	DC01
Active Directory Domain Services	Active Directory Domain Services
FSMO roles	FSMO roles
DNS	DNS
DHCP	DHCP
Group Policy	Group Policy
File services	File services
Security groups and permissions	Security groups and permissions
WS01	WS01
Linux01	Linux01
Linux VLAN placement	Linux VLAN placement
Active Directory integration	Active Directory integration
SSSD	SSSD
Kerberos	Kerberos
SSH	SSH
UFW	UFW
sudo controls	sudo controls
QEMU guest agent	QEMU guest agent
Centralized logging	Centralized logging
Backup architecture	Backup architecture
Snapshots	Snapshots
Archive integrity	Archive integrity
Restore testing	Restore testing
Storage monitoring	Storage monitoring
PowerShell automation	PowerShell automation
Read-only operations framework	Read-only operations framework
Authenticated target-local inventories	Authenticated target-local inventories
Troubleshooting and remediation	Troubleshooting and remediation
Known limitations	Known limitations

Topic	Document location
Skills demonstrated	Skills demonstrated
Conclusion	Conclusion
References	References
Appendices	Appendices

Table 7. Manual clickable TOC validation sample.

TOC entry tested	Result
Abstract	Hyperlink anchor present and page number populated.
Proxmox Virtualization	Hyperlink anchor present and page number populated.
Linux01	Hyperlink anchor present and page number populated.
Backup Architecture	Hyperlink anchor present and page number populated.
References	Hyperlink anchor present and page number populated.